SEAN (SUHYEON) YU

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SUMMARY

Master of Computer Science student at **Rice University** with experience building production web applications and an open-source library. Built an education web platform used by **5,000+ users**, developed an **interactive visualization dashboard** for machine unlearning, and shipped an **open-source library** for computing layer-wise similarity between neural networks.

EDUCATION

Rice University — Houston, TX, USA
Master of Computer Science, GPA: **3.91/4.00** — Aug. 2025 - Dec. 2026

Sungkyunkwan University (SKKU) — Top 3 University in South Korea — Suwon, South Korea
B.S. in Computer Science and Engineering, Senior Year GPA: **4.4/4.5** — Mar. 2018 - Aug. 2024

EXPERIENCE

Research Intern — Hybrid
Intelligent Data Computing Lab @ SKKU — Jun. 2024 – Aug. 2025

- Optimized scatterplot interaction **latency to p95 ≤ 11 ms at 2,000 points** via imperative **D3** updates and memoization.
- **Cut React re-renders by ~30%** by migrating global state from Context API to **Zustand** with selector-based subscriptions.
- Iterated on UI/UX 3 times in **Figma** based on user studies, refining interaction workflows to improve model evaluation.
- Deployed an open-source machine unlearning platform that garnered **~100 GitHub stars** and was published in **IEEE TVCG**.

Frontend Engineering Intern — Suwon, South Korea
Department of Computer Science and Engineering @ SKKU — Jan. 2023 – Jan. 2024

- Improved initial **load time by ~30%** through lazy loading and **TanStack Query**, and boosted SEO through Next.js SSR/SSG.
- Engineered the frontend with **Next.js** and TypeScript, scaling to **5,000+ users** and deploying via **Docker** and **AWS EC2**.
- Led **Agile** sprints and cross-functional team meetings, accelerating delivery and strengthening collaboration across teams.
- Re-architected the styling infrastructure from Styled Components to **Tailwind CSS**, eliminating CSS-in-JS runtime overhead.

PROJECTS

PyTorch-CKA | Open-Source Project — Dec. 2025 – Feb. 2026

- Accelerated CKA computation **44x** (ResNet-18/CIFAR-10) via vectorized **PyTorch** ops and concurrent GPU stream execution.
- Reduced GPU memory retention by **~96%** through batch-wise HSIC accumulation and explicit memory deallocation.
- Published the library to PyPI, achieving **~500 monthly downloads** and expanding accessibility of representation analysis.
- Simplified user workflow by enabling a single function call, reducing setup friction.
- Implemented visualization modules to render 2D model similarity matrices into customizable heatmaps and trend plots.

Decentralized Expense Settlement System | Start-up Project — Dec. 2022 – Feb. 2023

- Built the frontend in **React** and **JavaScript**, reducing component re-renders by **~80%** through strategic component splitting.
- Tracked **~100 active users** in Google Analytics and secured a competitive **~$3.5K startup grant**.
- Led end-to-end frontend development, taking full ownership of application architecture and UI design.

PUBLICATION

Jaeung Lee, **Suhyeon Yu**, et al, *Unlearning Comparator: A Visual Analytics System for Comparative Evaluation of Machine Unlearning Methods*, **IEEE Transactions on Visualization and Computer Graphics**, 2026. **(Impact Factor: 6.5)**

AWARDS

Dean's List & Merit-based Scholarship – **Top 3%** in CSE, SKKU — Apr. 2024, Mar. 2019
Gold Prize Winner, SKKU Hackathon — Sep. 2023

SKILLS

Full-stack Framework: Next.js, React Router| **Frontend**: React, Redux, HTML, CSS, D3 | **Backend**: Django, Node.js, REST APIs
Languages: JavaScript, TypeScript, Python, C/C++, SQL | **Cloud & DevOps**: AWS (S3, EC2, ECS, EMR, Lambda), CI/CD, Docker
Version Control: Git, GitHub | **Databases**: Supabase, Firebase, PostgreSQL, MySQL, MongoDB | **Unit Testing**: Vitest, Jest, pytest